

Can Talking to AI Chatbots Help When You're Feeling Down?

Examining the Role of Perceived Empathy

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Abstract— Mental health challenges are escalating globally, yet professional support remains critically limited. This research investigates whether AI chatbots can effectively facilitate emotion regulation and examines the role of perceived empathy in this process. In Study 1, participants ($N = 50$; 46 analyzed) recalled negative experiences and were randomly assigned to either a 20-minute chatbot conversation or a distraction task. Both conditions effectively reduced negative affect; however, only the chatbot condition additionally promoted positive affect and facilitated recovery to baseline levels. Following the interaction, participants rated the chatbot's dialogue-level empathic performance across three dimensions: understanding, responsiveness, and relational warmth. Results indicated that only the relational warmth significantly predicted emotion regulation outcomes, whereas perceived understanding and responsiveness did not. Study 2 employed a similar paradigm ($N = 60$) but manipulated conversational topic and chatbot prompt type, finding that regulatory efficacy remained stable across these variations. Participants retrospectively rated empathic quality at the sentence level. Results revealed that mean perceived empathy across turns predicted emotional outcomes, whereas empathy trajectory and variability did not. Together, the two studies position AI chatbots as promising, scalable tools for broadening access to emotional support, while offering concrete directions for designing conversational agents that prioritize empathic connection.

I. INTRODUCTION

Mental health has emerged as a critical global concern, with adolescent well-being declining and loneliness rising amid weakening social support networks and growing economic uncertainty [1], [2]. University students have been particularly affected, as depression, anxiety, self-harm, and suicidal ideation have increased markedly over the past decade [3]. Yet professional support remains severely limited—whether due to workforce shortages, lengthy waiting times, or persistent stigma surrounding help-seeking [4], [5], [6].

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These gaps have prompted growing interest in AI chatbots as accessible mental health tools; as of July 2025, roughly 5% of ChatGPT's 700 million users' conversations involve emotional or interpersonal concerns [7]. Yet whether AI chatbots effectively facilitate emotion regulation—and whether perceived empathy is the key underlying mechanism—remains empirically unresolved.

The present study addresses these questions through two objectives: (1) examining whether a 20-minute post-stressor chatbot conversation promotes emotion regulation, and whether effects generalize across different AI prompt conditions and emotional event types; and (2) investigating, at the dialogue level, how three dimensions of AI empathy (understanding, responsiveness, and relational warmth) differentially predict regulatory outcomes, and at the sentence level, whether the cumulative growth trajectory of perceived empathy across conversational turns shapes overall regulatory efficacy.

II. BACKGROUND AND RELATED WORK

A. Beneficial or Harmful? Examining AI's Role in Emotion Regulation

Evidence on AI chatbots and emotion regulation is mixed. On the positive side, AI systems can detect emotional cues on social media to predict depressive risk [8], [9]; chatbots like Wysa have demonstrated a therapeutic alliance with users and reduced anxiety and depressive symptoms [10]; and educational AI agents can recognize students' emotions and enhance their capacity for emotion regulation [11]. However, concerns are equally substantial. Rapid AI advancement has been linked to declining expectations for future well-being [12], workplace AI integration may increase stress and burnout [13], and over-reliance on AI risks reducing autonomy and social connection—contributing to anxiety [14], diminished meaning and motivation [15], and loneliness [16].

These mixed findings likely reflect substantial heterogeneity across studies in AI system type, interaction context, and emotional challenges examined, leaving it unclear whether mainstream AI chatbots can reliably support emotion regulation in everyday experiences. To address this gap, the present research systematically examines AI chatbot interactions under controlled conditions. Study 1 employs a negative memory recall paradigm to establish whether chatbot interactions regulate both negative and positive affect, compared to a distraction task. Building on these findings, Study 2 then tests whether such effects remain robust across different prompt types and communication topics.

B. The Role of Perceived AI Empathy in Emotion Regulation

Perceived empathy has been identified as a key mechanism underlying AI-assisted emotion regulation, yet its efficacy remains contested. Empirical evidence suggests that empathic AI features foster trust, enhance user experience, and deepen self-disclosure, ultimately strengthening emotional self-efficacy [17], [18], [19], [20] and AI-augmented empathic responses have even outperformed purely human responses in perceived helpfulness [21]. Nevertheless, skepticism persists: AI empathy may be fundamentally different from human empathy—primarily cognitive, and lacking affective and motivational depth [22]. Knowing that a response is AI-generated can reduce feelings of being understood [23], and empathic expressions that serve system designers' interests risk coming across as manipulative rather than genuine [24].

Given this debate, the present research further investigates the role of perceived AI empathy on emotion regulation. The two studies adopt different levels of empathy measurement to reflect their distinct aims. Study 1 employs a global dialogue-level perceived empathy scale as a moderator, providing an overall assessment of empathic quality following the interaction. Study 2 adopts sentence-level ratings to enable a more fine-grained analysis of how perceived empathy accumulates, evolves, and varies across conversational turns—capturing dynamic properties that global scales cannot detect and offering a more nuanced account of how empathic quality shapes regulatory outcomes.

III. STUDY 1: VALIDATION OF AI-ASSISTED EMOTION REGULATION AND MECHANISMS

A. Method

Fifty participants were recruited online and randomly assigned to a chatbot condition ($n = 25$) or a distraction control condition ($n = 25$). Four participants in the chatbot condition were excluded due to technical issues, yielding final samples of 21 (11 males, 10 females; $M_{\text{age}} = 22.57$, $SD = 2.99$) and 25 (11 males, 14 females; $M_{\text{age}} = 21.80$, $SD = 1.85$), respectively. All participants in two studies provided informed consent under NTU Research Ethics Committee approval (NTU-REC-202409HS026). A debriefing session was conducted at the conclusion of each study to address any emotional concerns arising from the procedure.

The procedure is detailed in Figure 1. Following a baseline affect assessment, participants came to the laboratory and completed an 8-minute negative memory-recall task to induce negative affect, then engaged in a 20-minute task. Participants in the chatbot condition interacted with a ChatGPT-4o-based chatbot prompted to act as an empathic listener, and were instructed to talk with the chatbot about the negative experience they had just recalled. Participants in the

distraction control condition completed a classic Stroop task, a color–word identification task.

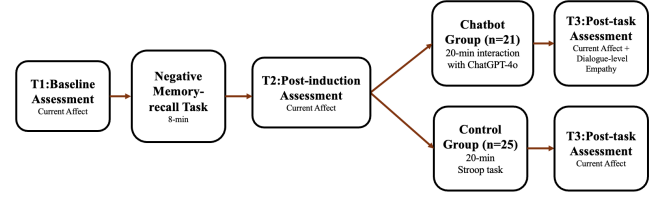


Figure 1. Experimental Procedure of Study 1

Affect was measured using the PANAS (10-item Positive Affect Scale and Negative Affect Scale; 5-point scale); [25] at baseline (T1), post-induction (T2), and post-interaction (T3). At the post-interaction test, the chatbot's dialogue-level empathy was assessed using the Chinese version of perceived robot empathy scale, adapted from previous measures of robot empathy [26], [27]. The scale comprises three dimensions: Empathic Understanding, reflecting the perceived ability to understand others' feelings (e.g., "The chatbot cares about my feelings"); Empathic Response, capturing the perceived ability to express supportive behaviors (e.g., "The chatbot encourages me"); and Empathic Relationship, emphasizing participants' sense of comfort and ease during the interaction (e.g., "I felt at ease during my interaction with the chatbot."). To validate the Chinese version of the scale, a pilot study was conducted to evaluate its psychometric properties in a separate sample of 230 participants in an online chatbot interaction context. The instrument demonstrated high internal consistency (Cronbach's $\alpha = .91$), and confirmatory factor analysis (CFA) supported its construct validity with excellent model fit indices (CFI = .99, TLI = .99, RMSEA = .01, and SRMR = .02), indicating strong validity in chatbot interaction contexts.

B. Results

The negative memory-recall task successfully induced negative affect and reduced positive affect, with no significant between-group differences in the degree of emotional induction ($ps > .656$), indicating that our negative event recall manipulation was highly effective (See Figure 2; T1-T2).

For negative affect, both the chatbot group ($b = .876$, $p < .001$) and the control group ($b = .514$, $p = .003$) showed significant reductions following the intervention (See Figure 2, T2-T3); however, the magnitude of reduction did not differ significantly between groups ($b = .362$, $p = .104$). For positive affect, a significant interaction emerged ($b = .357$, $p = .036$): the chatbot group recovered significantly following the interaction ($b = 0.48$, $p < .001$), whereas the control group did not ($b = 0.12$, $p = .273$). Furthermore, the chatbot group's post-interaction positive affect did not differ significantly from baseline ($b = 0.18$, $p = .133$), whereas the control group's post-interaction positive affect remained significantly below baseline ($b = -0.281$, $p = .037$). At the level of individual positive affect items, only the chatbot group showed significantly greater increases relative to the control group on three specific items—Strong, Inspired, and Interested ($ps < .05$).

—suggesting that the chatbot group's enhancement of positive affect was selective rather than global.

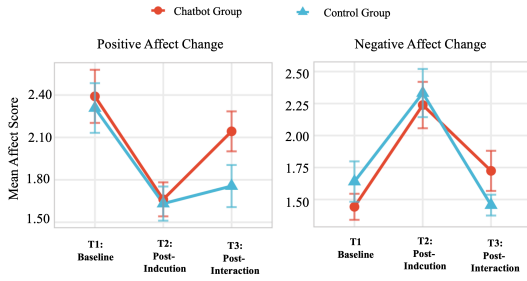


Figure 2. Mean positive and negative affect scores across three time points (T1: Baseline, T2: Post-Induction, T3: Post-Interaction) for the Chatbot and Control groups.

Regarding the main effect of dialogue-level Empathy, the composite empathy scores marginally significantly predicted positive affect recovery ($b = 0.23$, $p = .079$) and exerted a significant effect on negative affect reduction ($b = -0.35$, $p = .031$). Among the three dialogue-level empathy dimensions, only Empathic Relationship significantly moderated both the improvement in positive affect ($b = 0.38$, $p = .001$) and the reduction in negative affect ($b = -0.41$, $p = .011$), whereas Empathic Understanding and Empathic Response were non-significant predictors of either positive affect improvement or negative affect reduction ($ps > .05$).

C. Contextual Limitations and Transition to Study 2

Study 1 establishes that, relative to the control condition (traditional distraction strategy), AI chatbot interaction significantly facilitated the recovery of positive affect regulation, with this facilitative effect concentrated specifically on the emotions of Strong, Inspired, and Interested—three affective states that have been linked to psychological resilience [28]. However, the current design leaves open the question of boundary conditions: does AI-mediated emotion regulation generalize across different conversational contexts and discussion topics? Study 2 addresses both issues directly.

The findings from Study 1 further indicate that, among the three dimensions of dialogue-level perceived empathy—Empathic Understanding, Empathic Response, and Empathic Relationship—only Empathic Relationship significantly predicted affective recovery in the chatbot group. This suggests that how participants felt during the interaction mattered more than their perceptions of the chatbot's technical capabilities. Future chatbot design might therefore prioritize the quality of users' felt experience—rather than solely the system's technical sophistication. Moreover, the dialogue-level empathy measure employed in Study 1 cannot reveal how empathic moments accumulate over the course of a conversation, nor whether it is the sustained quality, the trajectory, or the variability of AI empathy that drives regulatory outcomes. To address this limitation, Study 2 replaced the global empathy rating with a sentence-level retrospective evaluation, enabling analysis of empathy trajectories, variability, and accumulation as predictors of emotional improvement.

IV. STUDY 2: CONTEXTUAL STABILITY AND GRANULAR MECHANISMS

A. Method

Study 2 recruited an independent sample of 60 participants (25 males, 35 females; $M_{age} = 22.18$, $SD = 1.93$) using a mixed-design approach in which each participant interacted with both a standard, unprompted ChatGPT-4o (control condition) and one of two prompted versions. Participants were randomly assigned to either the Tone condition, where the AI adopted the colloquial conversational style of Taiwanese college students to enhance perceived relational naturalness [29]—or the Similarity condition—where the AI mirrored participants' personal profiles (e.g., major, personality traits) to test whether perceived interpersonal similarity enhances regulatory outcomes [30]. The overall procedure closely paralleled that of Study 1: participants first recalled a negative emotional experience for eight minutes, followed by a 20-minute conversation with the chatbot.

To capture contextual variability in emotion regulation, two categories of negative emotional experiences were introduced as conversation topics: interpersonal issues (e.g., family or peer conflicts) and career issues (e.g., academic or career stress). A counterbalanced design was implemented to control for order and topic effects: each participant discussed one topic category with the control model and the other with the prompted model, with sequence counterbalanced across participants. Unlike the pre-post affect measurement in Study 1, emotional improvement was quantified using a single 9-point retrospective scale administered after each interaction (1 = significantly worsened, 5 = no change, 9 = significantly improved). To capture the dynamic nature of sentence-level empathy throughout the conversation, participants also completed a segment-by-segment retrospective rating at the end of each session, evaluating the empathic quality of each AI-generated response on a 5-point Likert scale (see Figure 3).

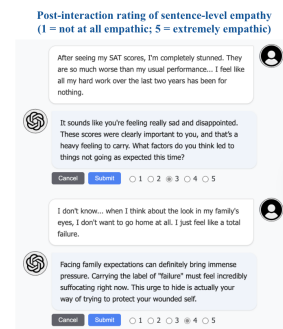


Figure 3. Dynamic measurement of chatbot's sentence-level empathy.

B. Results

The analysis first examined the general efficacy of AI-mediated emotion regulation across different conditions (See Figure 4). One-sample t-tests against the neutral point ($\mu = 5$) indicated that all three prompt conditions—Control ($t(59) = 6.56$, $p < .001$), Tone ($t(29) = 4.10$, $p < .001$), and Similarity ($t(29) = 7.35$, $p < .001$)—produced significant emotional improvement. Despite these robust effects, no significant

differences were found between the prompted and unprompted versions, suggesting that the baseline capability of large language models for emotional support is already highly effective. Furthermore, chatbots demonstrated significant emotion-regulation capabilities across both career ($t(59) = 8.81, p < .001$) and interpersonal ($t(59) = 5.77, p < .001$) conversation topics.

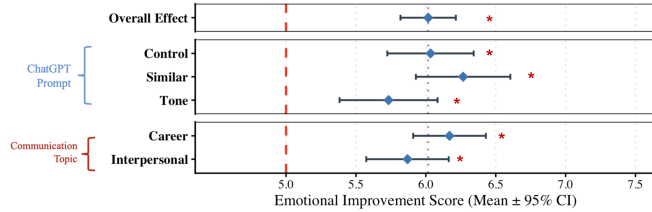


Figure 4. Stability of Emotion Regulation Effects Across Prompts and Communication Contexts.

The investigation into the dynamic mechanisms of sentence-level empathy revealed that the overall average empathy score was the primary predictor of emotional recovery (See Figure 5). A partial correlation analysis demonstrated a stable, positive relationship between average perceived empathy and emotional improvement ($b = 0.25, p = .006$). In contrast, the temporal trajectory of empathy (e.g., whether empathy increased over the course of the conversation) and its variability showed no significant association with the emotional improvement ($b = 0.05, p = .588; b = -0.03, p = .776$, respectively).

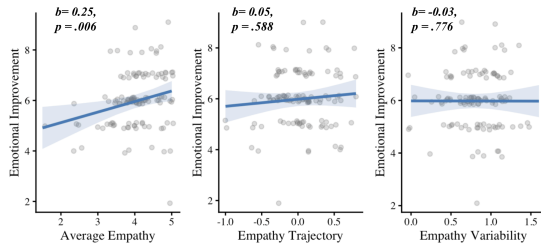


Figure 5. Effects of Sentence-level Empathy on Emotional Improvement

V. DISCUSSION

A. AI-Assisted Emotion Regulation During Negative Affect

Study 1 demonstrated that, compared to a traditional distraction task, a 20-minute AI chatbot interaction more effectively facilitated positive emotion regulation, returning emotional states to baseline. Study 2 replicated and extended these findings, showing regulatory stability across diverse prompts and conversational topics. These results align with prior evidence supporting AI-driven emotional interventions [10], [11], suggesting that AI chatbots may serve as a scalable, cost-effective resource for managing daily stressors—particularly for those who face elevated depression risk yet lack access to professional counseling [31]. Their immediate availability may also complement traditional strategies such as expressive writing [32] and help prevent rumination-driven emotional escalation [33]. Notably, chatbot interactions specifically enhanced high-activation positive emotions linked

to psychological resilience [28]—such as feeling strong, inspired, and interested—positioning AI chatbots as a promising resilience-building tool that helps individuals reconstruct positive experiences in adversity.

B. Dialogue- and Sentence-Level Empathy in AI-Assisted Emotion Regulation

At the dialogue level, only the Relationship dimension of empathy—emphasizing positive emotional connection—significantly predicted regulation outcomes, while Understanding and Response dimensions did not. This mirrors evidence that emotional, rather than cognitive, support is the primary driver of cognitive flexibility under stress [34]. At the sentence level, mean empathy per conversational turn, rather than trajectory or variability, significantly predicted efficacy. Unlike human therapy, where occasional empathic lapses may be tolerated, AI-mediated support appears to require consistent quality across every turn. Future designs should adopt active-constructive responding strategies to maximize the emotional value of each interaction [35].

C. Limitation

Participants were drawn from the general population; generalization to clinical populations (e.g., major depressive disorder, bipolar disorder) requires caution, as AI may fail to detect crisis signals or provide adequate clinical judgment. Future work should establish safety protocols in collaboration with clinical experts.

VI. CONCLUSION

Two studies consistently demonstrated that AI chatbots possess meaningful emotion-regulation capacity, robust across different prompt types and conversational topics. The relational dimension of empathy emerged as a critical determinant, and mean perceived empathy per turn—rather than its trajectory or variability—was the key predictor of outcomes. These findings suggest that emotionally oriented AI chatbots hold significant potential to expand accessible regulatory strategies. Future design should prioritize empathy at the turn level, focusing on how each response fosters genuine comfort to maximize regulatory efficacy.

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DECLARATION OF RAW DATA AND AI USE

All raw data and analysis code are available on OSF (https://osf.io/rxukp/overview?view_only=5c4e5f73370145faaeca0caecda529a3). ChatGPT was used for English language polishing only; all intellectual content was developed by the authors.

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